Traceability Supplement

TABLE XIV
PROPOSED, UNMEASURED SOURCE-DERIVED TRACEABILITY DIMENSIONS

ID	Required correspondence	Unit numerator / denominator
TR01	identity/activity binding to task, exact artifact revision/digest, activity, and actor	complete slots / required slots
TR02	request/input → rule version → evaluated check/result → decision	complete chains / inventoried decisions
TR03	decision → observed state or artifact effect with before/after evidence	complete correspondences / decisions requiring outcome
TR04	triggering failure → corrective package → exact revalidation → result	complete repair chains / actual repair attempts
TR05	forward requirement → artifact → verification and reverse in-scope change mapping	report forward and reverse separately
TR06	non-author independent exercise of declared audit/replay target	exercised targets / declared targets

The instrument is proposed and unmeasured; no score was collected or validated. Data and Code Availability identifies the codebooks and source rationale.

I. PROPOSED TRACEABILITY INSTRUMENT (NOT MEASURED)

The proposed instrument assesses correspondence across requirements, artifacts, actions, and outcomes rather than correctness, hidden reasoning, developer skill, or player benefit. For each dimension d, the protocol freezes an arm-neutral inventory U_d and the required links $L_d(u)$ for each unit u, letting N_d count applicable units in U_d and n_d count units with every required link verified. Independent assessors award 1 only when the exact source and version, raw-evidence relations, and locator are all verified; otherwise they award 0 and classify the gap as missing, inaccessible, stale, contradicted, or unverified evidence. The ratio n_d/N_d is reported only when the inventory is complete and $N_d > 0$, alongside assessor disagreement and seeded-corruption controls, without forming weighted composites. Incomplete inventories are marked INDETERMINATE together with observed counts and the inventory gap, and an N/A designation requires independent evidence of non-applicability. Independently verified zero actual repair attempts can qualify a repair item, whereas an empty log cannot. Finally, PASS, FAIL, NOT_RUN, STAGE_NOT_REACHED, OUT_OF_SCOPE, and MISSING must remain distinct, and `stage_not_reached` never counts as a pass.